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OCR GCSE Geography B (J384) — Paper 1: Our Natural World · Topic 4: Sustaining Ecosystems

1.7 Sustaining Ecosystems: Polar Environments

1. Where Are Cold Environments Found?

Cold environments — areas where temperatures are very low for most of the year, so ice, snow and frozen ground dominate. Geographers divide them into **polar** environments (Antarctica, the Arctic Ocean ice and Greenland ice sheet) and **tundra** environments (the treeless lands of northern Canada, Alaska, northern Russia and Scandinavia that fringe the Arctic). Some high mountain areas (alpine) are also cold environments.

Feature	Polar	Tundra
Location	Within the Arctic and Antarctic Circles (66.5° N and S) to the poles	Roughly 60°–75° N, between the polar ice and the coniferous forest (taiga)
Temperature	Winter as low as −50°C (Antarctica's record is −89.2°C); summer rarely above 0°C	Winter −30°C to −40°C; brief summer of 6–10 weeks reaching 5–15°C
Precipitation	Very low — under 250 mm a year, mostly snow (polar deserts)	Low — around 150–380 mm a year, mostly snow, some summer rain
Ground	Permanently covered by ice sheets and glaciers	Permafrost beneath a thin active layer that thaws each summer
Vegetation	Almost none — some mosses and lichens on bare rock	Low-growing mosses, lichens, grasses, sedges and dwarf shrubs; no trees

2. Physical Characteristics of a Cold Environment

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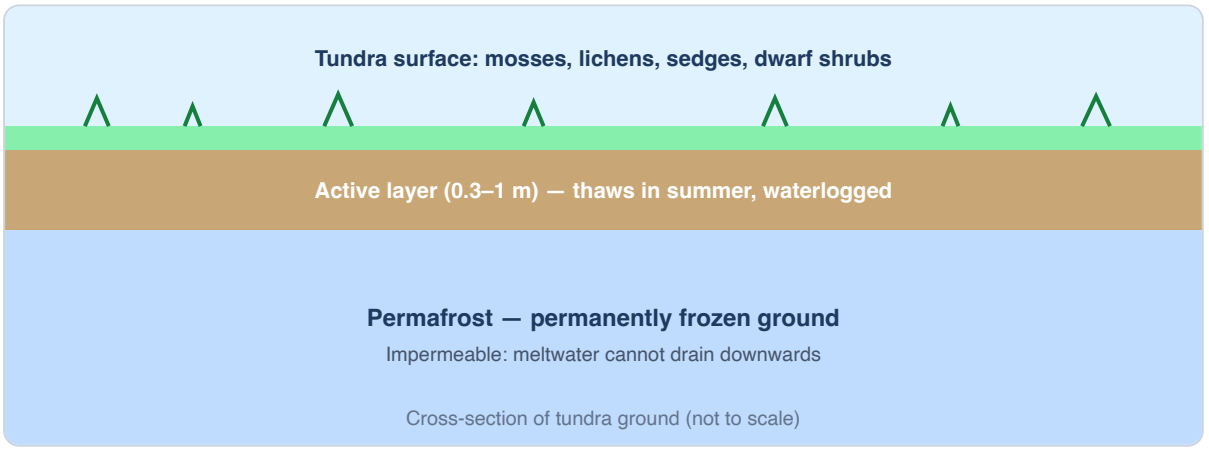
Climate

- High latitude means the sun's rays strike the ground at a low angle and pass through more atmosphere, so they are spread over a larger area and heating is weak.
- Snow and ice have a **high albedo** — they reflect up to 80–90% of incoming sunlight, keeping temperatures low.
- Cold, sinking air creates high pressure, so precipitation is low. Strong katabatic winds blow off the ice sheets.
- Extreme seasonal variation in daylight — the "midnight sun" in summer and 24-hour darkness in winter.

Permafrost and soils

> **Permafrost** — ground that remains frozen (at or below 0°C) for at least two consecutive years. In the tundra it can be hundreds of metres deep. Above it is the **active layer**, which thaws in summer.

- Because meltwater cannot drain through the frozen ground below, the active layer becomes **waterlogged** in summer, creating bogs, marshes and shallow lakes.
- Soils are thin, acidic and infertile. Low temperatures mean decomposition is extremely slow, so dead plant material builds up as **peat** — a huge store of carbon.



Cross-section of tundra ground (not to scale)

3. Interdependence in Cold Environments

- **Climate → plants:** the short growing season (6–10 weeks) and low temperatures limit plants to small, slow-growing species.

- **Permafrost → plants:** roots cannot penetrate the frozen ground, so trees cannot grow; waterlogging favours mosses and sedges.
- **Plants → animals:** lichens and mosses are the main food for caribou (reindeer) and lemmings; lemmings are prey for Arctic foxes and snowy owls.
- **Plants → permafrost:** the vegetation and peat layer **insulate** the ground, protecting the permafrost from summer heat. If vegetation is removed (by vehicles or buildings), the permafrost thaws, the ground subsides and becomes boggy.
- **Animals → people:** Indigenous peoples such as the Inuit and Iñupiat have traditionally depended on caribou, seals, whales and fish for food, clothing and tools.

4. Adaptations of Plants and Animals

Plants	Animals
Low-growing and cushion-shaped (e.g. moss campion) — sheltered from strong, cold winds and insulated by snow in winter	**Thick fur or feathers and fat (blubber)** — insulation against extreme cold (polar bear, seal, musk ox)
Shallow roots — only the active layer thaws, so roots spread sideways	**White coats** for camouflage in snow; Arctic fox and Arctic hare change to brown in summer
Short life cycles — flower and set seed quickly in the brief summer (e.g. Arctic poppy)	**Small extremities** (short ears, tails, legs) reduce heat loss (Arctic fox)
Dark leaves absorb more of the weak sunlight; some, like the Arctic poppy, turn their flowers to follow the sun	**Migration** — caribou migrate up to 5,000 km a year; many birds visit only in summer
Small, waxy or hairy leaves reduce water loss in cold, dry, windy conditions (bearberry)	**Hibernation** or burrowing under snow (Arctic ground squirrel, lemmings)

5. Biodiversity

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- Cold environments have **low biodiversity** — only around 1,700 plant species live in the Arctic tundra, compared with tens of thousands in the Amazon.
- Food webs are **simple**, with few species at each level. A change to one species (e.g. a crash in lemming numbers) can have major knock-on effects — the ecosystem is **fragile**.
- **Issues:** climate change (the Arctic is warming nearly four times faster than the global average), loss of sea ice for polar bears, oil spills, and the spread of species from further south.

6. Case Study: Alaska — Opportunities and Challenges

Alaska, USA

Alaska is the largest US state (about 1.7 million km²), lying mainly between 55°N and 71°N. Its north is Arctic tundra underlain by permafrost; the population is only about 740,000, mostly in the south around Anchorage. About 15% of residents are Alaska Native peoples.

Development opportunities

- **Oil and gas:** the Prudhoe Bay oilfield on the North Slope, discovered in 1968, was the largest in North America. Oil revenue funds most of the state's budget and the **Alaska Permanent Fund**, which pays every resident an annual dividend (typically $1,000–$2,000).
- **Mineral extraction:** gold, zinc and lead — the Red Dog Mine is one of the world's largest zinc mines.
- **Fishing:** Alaska lands around 60% of all US seafood (salmon, pollock, crab), employing tens of thousands of people.
- **Tourism:** over 2 million visitors a year come for wilderness, glaciers, whale-watching and the Northern Lights — cruise ships along the coast and national parks such as Denali.

Development challenges

- **Extreme temperatures:** winter temperatures below −40°C make outdoor work dangerous, freeze machinery and require heated buildings, adding cost.
- **Inaccessibility:** vast distances, mountains and few roads. Many communities can only be reached by air or by sea; ice roads are used in winter.
- **Buildings and infrastructure:** heat from buildings and pipes can thaw permafrost, making the ground subside and buildings crack. Buildings are raised on stilts or piles, and the **Trans-**

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Alaska Pipeline (1,287 km, from Prudhoe Bay to the ice-free port of Valdez) is raised on supports for about half its length, with heat pipes to keep the ground frozen, and built in a zigzag to cope with ground movement and earthquakes.

- **Environmental risk:** the Exxon Valdez oil spill (1989) released around 41 million litres of oil into Prince William Sound, killing hundreds of thousands of seabirds and thousands of sea otters.

7. The Value of Cold Environments as Wilderness Areas

Wilderness — a natural area largely undisturbed by human activity, with few or no permanent inhabitants, roads or buildings.

- **Scientific research:** ice cores from Antarctica and Greenland hold a record of past climates going back 800,000 years; cold environments are an early warning of climate change.
- **Biodiversity:** unique species found nowhere else (polar bears, emperor penguins, musk oxen).
- **Global climate regulation:** high-albedo ice reflects sunlight; permafrost stores around 1,500 billion tonnes of carbon — roughly twice as much as the atmosphere.
- **Indigenous cultures** and traditional ways of life.
- **Unspoilt landscapes** valued for their beauty, recreation and wellbeing.

Why are cold environments fragile?

- Plants grow extremely slowly, so damaged vegetation can take decades to recover — vehicle tracks on tundra can remain visible for 50 years or more.
- Low temperatures mean oil and waste break down very slowly.
- Simple food webs mean the loss of one species affects many others.
- Thawing permafrost releases methane and CO_2, causing further warming (a positive feedback).

8. Balancing Development and Conservation

Strategy	How it works	Example and evaluation

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Use of technology	Reduce the footprint of development: raised buildings and pipelines, horizontal drilling from a single pad, ice roads that melt away in summer	Trans-Alaska Pipeline allows caribou to pass underneath; drilling pads now cover a fraction of the area they did in the 1970s
Role of governments	Laws and protected areas restrict where development can happen and set environmental standards	Arctic National Wildlife Refuge (ANWR) protects 78,000 km²; but drilling leases in its coastal plain have been debated for decades and remain politically contested
International agreements	Countries agree to limit activity in shared environments	The **Antarctic Treaty** (1959) reserves Antarctica for peaceful scientific research; its 1991 Madrid Protocol bans mining. The **Arctic Council** coordinates eight Arctic nations on environmental protection
Conservation groups	NGOs campaign, monitor and raise awareness, lobbying governments and companies	WWF and Greenpeace campaigns against Arctic oil drilling; Greenpeace's "Save the Arctic" campaign contributed to Shell abandoning Arctic drilling in 2015

Exam Tip: A 9-mark question might ask "To what extent can development and conservation be balanced in cold environments?" Use Alaska: technology (raised pipeline) and laws (ANWR) show some balance, but oil spills and permafrost thaw show the risks remain. Reach a judgement — for example, that balance depends on strict regulation and on how much the local economy depends on extraction.

9. Antarctica — Characteristics and Threats

- **Location and size:** a continent of about 14 million km² centred on the South Pole, almost entirely covered by an ice sheet averaging over 2 km thick — around 70% of the world's fresh water.
- **Climate:** the coldest, driest and windiest continent; the interior is a polar desert with under 50 mm of precipitation a year. The lowest temperature recorded on Earth (−89.2°C) was at Vostok station.

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- **Wildlife:** almost all life is found on the coasts and in the Southern Ocean — penguins, seals, whales, seabirds and krill, which is the base of the marine food web.
- **People:** no permanent population; around 1,000–5,000 scientists and support staff at research stations depending on the season.

How people use Antarctica

- **Scientific research** — ice cores, climate monitoring, ozone layer studies (the ozone hole was discovered by the British Antarctic Survey in 1985).
- **Tourism** — over 100,000 visitors a season, mostly on cruise ships to the Antarctic Peninsula.
- **Fishing** — krill and toothfish in the Southern Ocean.

Threats

- **Climate change:** the Antarctic Peninsula is one of the fastest-warming places on Earth; ice shelves such as Larsen B (2002) have collapsed; sea ice extent hit record lows in 2023.
- **Tourism:** risk of oil spills from ships, disturbance to wildlife, trampling and introduction of non-native species.
- **Overfishing:** of krill and toothfish, affecting whales, seals and penguins.
- **Possible future resource exploitation** — minerals and oil, if the mining ban were ever lifted.

10. Managing Antarctica Sustainably (Global Scale)

Strategy	How it works	Evaluation
Antarctic Treaty (1959)	Signed by 12 countries, now over 50; reserves Antarctica for peace and science; territorial claims are frozen	Remarkably successful for over 60 years — no military activity, strong scientific cooperation
Madrid Protocol (1991)	Bans mining and sets strict environmental rules for waste and research stations	Mining ban can only be reviewed after 2048; relies on continued agreement

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IAATO tourism guidelines	International Association of Antarctica Tour Operators limits ships carrying over 500 passengers from landing; no more than 100 people ashore at a site at once; keep 5 m from wildlife	Self-regulated by the industry; growing visitor numbers increase pressure
CCAMLR (Commission for the Conservation of Antarctic Marine Living Resources)	Sets catch limits for krill and toothfish; created the Ross Sea Marine Protected Area (2016), one of the world's largest	Agreement on new protected areas is slow because decisions need consensus

Exam Tip: OCR asks you to compare sustainable management at a *small scale* in one polar region (e.g. Alaska or Svalbard in the Arctic) with *global-scale* management (the Antarctic Treaty). Show why management is easier in Antarctica — no permanent population, no economy to protect — than in the Arctic, where millions of people live and countries have sovereign rights.

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